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Version History

Version Name	Specifics	Last Updated On
Version 1.0	Initial Draft of Kerala Mart SRS	2026-02-13

1. Executive Summary

1.1 Purpose

This document defines the complete functional and non-functional requirements for Kerala Mart, a multivendor marketplace platform where buyers browse products and directly chat with sellers. The platform does not process buyer payments. Sellers must purchase a subscription to list products.

1.2 Project Scope

The scope of the Kerala Mart project encompasses the design, development, deployment, and maintenance of a web-based multivendor marketplace platform that connects buyers and verified sellers within Kerala and potentially across India.

Kerala Mart will function as a listing-based digital marketplace, where sellers can showcase their products and buyers can directly communicate with sellers to complete transactions offline. The platform will not facilitate buyer-side payment processing or checkout functionality. Instead, it will act as a communication and discovery platform.

1.3 Project Success Matrix

The Project Success Matrix for the Kerala Mart Marketplace Development outlines the key criteria that define the success of the project. The matrix includes both quantitative and qualitative measures, focusing on platform growth, user engagement, revenue sustainability, system performance, and regulatory compliance.

- **Seller Onboarding:**

Criteria: Seller Onboarding Growth Rate

Measurement: Track the number of new seller registrations and approved GST-verified sellers on a monthly basis. Monitor the conversion rate from registered sellers to active subscribed sellers. Evaluate subscription renewal rate and seller churn rate over time.

Measure the average approval turnaround time for GST verification.

- **Buyer Engagement:**

Criteria: Buyer Chat Engagement Rate

Measurement: Monitor the number of product detail page visits compared to the number of chat conversations initiated. Track the percentage of product views that result in chat initiation. Measure average number of messages exchanged per chat session and seller response time. Analyse repeat buyer engagement and daily active chat users.

- **Subscription Revenue:**

Criteria: Subscription Revenue Growth and Stability

Measurement: Track the total number of active seller subscriptions. Monitor Monthly Recurring Revenue (MRR) and Annual Recurring Revenue (ARR). Evaluate subscription renewal rate, churn rate, and distribution of subscription plans (Monthly, Quarterly, Annual). Compare revenue growth trends across reporting periods.

- **Platform Performance:**

Criteria: System Response Time and Uptime

Measurement: Conduct regular performance testing to ensure page load time remains under defined thresholds. Monitor system uptime percentage and server performance under peak load conditions. Track frequency of system errors and downtime incidents.

- **Security & Compliance:**

Criteria: Security Audit Results and GST Verification Accuracy

Measurement: Perform periodic security audits and vulnerability assessments. Monitor GST validation success rate and detect fraudulent registrations. Track unauthorized access attempts and ensure encrypted communication protocols are implemented across the platform.

- **User Satisfaction:**

Criteria: Buyer and Seller Satisfaction Scores

Measurement: Conduct periodic surveys and collect feedback from buyers and sellers. Track complaint resolution time and platform usability ratings. Monitor subscription feedback and seller retention rates.

- **Project Timeline:**

Criteria: Adherence to Development Milestones

Measurement: Compare planned development milestones with actual completion dates. Monitor sprint velocity and track delays or scope deviations.

- **Budget Adherence:**

Criteria: Budget Compliance

Measurement: Perform financial reviews to ensure development and operational costs remain within the approved budget. Track infrastructure expenses and subscription revenue against projections.

- **Stakeholder Satisfaction:**

Criteria: Stakeholder Feedback and Approval

Measurement: Conduct regular stakeholder review meetings. Document feedback and track approval at each major milestone of development.

2. System Overview

2.1 Proposed System

The proposed Kerala Mart system is a web-based multivendor digital marketplace platform designed to connect buyers and verified sellers through a structured listing and communication model. Unlike traditional e-commerce platforms, Kerala Mart does not facilitate online checkout or payment processing for buyers. Instead, it functions as a product discovery and communication platform where transactions are completed offline between buyers and sellers.

The platform is built on a subscription-based seller onboarding model and a GST-compliant verification process to ensure business authenticity and regulatory compliance.

2.2 Key Features & Capabilities

The following are the key features and capabilities of the proposed Kerala Mart Web Application.

- **Separate Buyer & Seller Login:**
The platform shall provide two distinct authentication modules — one for Buyers and one for Sellers. Each user role shall have separate dashboards and access privileges. Role-based access control shall ensure that users can only access features relevant to their account type.
- **Buyer Registration & Login:**
Seamless account creation and login for buyers. Buyers shall be able to browse products without login; however, login shall be mandatory to initiate chat with sellers.
- **Seller Registration with GST Verification:**
Sellers shall be required to register by providing mandatory business details including GST number. GST format validation shall be enforced. Sellers must upload GST certificate documentation. Seller accounts shall remain inactive until approved by Admin after verification.
- **Subscription-Based Product Listing:**
The system shall operate on a subscription model. Sellers must purchase an active

subscription plan (Monthly / Quarterly / Annual) before listing products. Product listing capability shall automatically deactivate upon subscription expiry.

- **Seller Dashboard:**
A comprehensive dashboard shall be provided for sellers to manage products, view subscription status, respond to chat inquiries, and update profile details.
- **Product Browsing & Filtering:**
The platform shall provide intuitive browsing functionality including category-based listing, keyword search, and filtering options such as price range and seller. Products shall be displayed in structured listing format with relevant details.
- **Product Listing & Detail Page:**
Each product shall have a dedicated detail page displaying product images, description, price (display only), seller details, and a “Chat with Seller” button.
- **Direct Buyer-Seller Chat:**
The platform shall include a secure real-time chat system enabling direct communication between buyers and sellers. Chat history shall be stored, and notifications shall be provided for new messages. This feature replaces the traditional cart and checkout model.
- **Admin Moderation Dashboard:**
A centralized Admin panel shall be provided to manage sellers, verify GST details, configure subscription plans, moderate product listings, manage users, monitor chat activity, and generate analytical reports.
- **Content Management:**
Admin shall be able to manage homepage banners, categories, featured listings, and legal pages dynamically from the backend.
- **Role-Based Access Control:**
The system shall implement role-based permission management for Buyers, Sellers, and Admin users to ensure secure access to system functionalities.

- **Secure & Scalable Architecture:**
The platform shall be built using secure authentication mechanisms, encrypted communication (SSL), and scalable infrastructure capable of handling concurrent users.

2.3 Target Audience

The target audience for Kerala Mart consists of both business sellers and end consumers who seek a secure, GST-verified, communication-driven marketplace platform. The audience can be categorized as follows:

- **Local Small & Medium Businesses (SMEs):**
Small and medium-scale businesses operating within Kerala who wish to expand their market presence through a digital platform. These businesses may include retail shops, wholesalers, manufacturers, home-based entrepreneurs, and service providers. Kerala Mart provides them with a low-entry digital marketplace through subscription-based listing without the complexity of managing payment gateways or logistics systems.
- **Home-Based Sellers & Entrepreneurs:**
Individuals running home-based businesses such as handicrafts, clothing, food products, or specialty goods. Kerala Mart offers them an opportunity to list products and connect directly with buyers through chat-based negotiation without technical complexities.
- **Retail Buyers & Consumers:**
Individual customers looking to browse and discover products from local verified sellers. Buyers who prefer direct communication and negotiation with sellers rather than automated checkout systems will benefit from the chat-based interaction model.
- **Local Community Shoppers:**
Customers who prefer purchasing from Kerala-based businesses and supporting local sellers. The platform encourages regional commerce by connecting local buyers with verified local businesses.

- **Restaurants & Food Businesses:**
Restaurants, cloud kitchens, catering services, bakeries, and food entrepreneurs who wish to display their menu items, promotional offers, or specialty dishes. Customers can browse listings and contact restaurants directly through chat to place orders or make inquiries. Transactions and deliveries are managed offline by the restaurant.

3. Functional Requirements

3.1.1 Homepage

3.1.1 Homepage

The users shall be displayed with a Homepage upon landing on the Kerala Mart website. Kerala Mart is a web-based e-commerce listing platform focused on vegetables, fruits, and spices. The Homepage shall serve as the primary entry point for users and shall be fully customizable from the backend, allowing the Admin to manage, rearrange, enable, or disable sections dynamically.

The Homepage of the website shall include the following components:

- **Logo:**

The Kerala Mart logo shall be displayed prominently at the top of the website. Clicking the logo shall redirect the user to the Homepage from any other page within the website.

- **Notification Bar:**

A dynamic notification bar shall be displayed at the top of the website to show announcements such as:

- Daily vegetable price updates
- Seasonal availability
- Promotional offers
- Important notices

The content of this notification bar shall be configurable from the Admin panel.

- **Header Navigation:**

The website shall include a header navigation menu providing access to:

- Home
- Categories
- Buyer Login / Register
- Seller Login
- Sell on Kerala Mart (CTA)
- Contact Us

The header shall remain consistent across all website pages.

● **Category Navigation Section:**

The Homepage shall display major product categories such as:

- Fresh Vegetables
- Fresh Fruits
- Spices
- Organic Produce

Each category shall redirect users to the respective Product Listing Page.

● **Search Bar:**

A search functionality shall allow users to search for specific vegetables, fruits, spices, or seller names. Search results shall be displayed on the Product Listing Page.

● **Featured Listings Section:**

The Homepage shall display featured products selected by Admin. Each listing shall include:

- Product Image
- Product Name
- Price (Display Only – e.g., Per Kg)
- Seller Name
- Seller Location
- “Chat with Seller” Button

The website shall not include “Add to Cart” or “Buy Now”

● **Daily Price Highlights:**

A section may display updated market prices for commonly traded vegetables, fruits, and spices.

● **Verified Sellers Section:**

A section highlighting GST-verified sellers shall be displayed to build buyer trust.

● **Footer Section:**

The footer shall include:

- About Us
- Seller Registration
- Privacy Policy
- Terms & Conditions
- Contact Information
- Support Details

The footer content shall be manageable from the Admin panel.

● **Responsive Web Design:**

The website shall be fully responsive and optimized for desktop and mobile browsers.

● **Security:**

The website shall operate over HTTPS protocol and implement secure authentication mechanisms.

Story ID	KM001
User Story Name	Homepage
As a	Buyer ,Seller
I want to	View the homepage
So that	I can browse products and navigate through the platform
Pre-Conditions	• The user has entered the Kerala Mart website URL in the browser • The system server is active and operational. • The user has an active internet connection.
Acceptance Criteria	• The system shall display the Kerala Mart logo prominently on the homepage. • Clicking the logo shall redirect the user to the homepage from any page. • The header shall include Buyer Login and Seller Login options.

	• The system shall display a search bar for product lookup. • The system shall display product categories for easy navigation. • The system shall display featured products and recently added products (if enabled). • Each product card shall include Product Image, Product Name, Price (display only), Product Name, and “Chat with Seller” button. • The homepage shall be accessible without mandatory login. • The system shall enforce role-based access control for restricted actions. • The system shall log homepage visit activity for analytics and monitoring. • The system shall ensure secure communication using SSL encryption. • The homepage shall load within acceptable performance thresholds.
Alternate Flow	If the server is unavailable, the system shall display a maintenance page. • If product data fails to load, a user-friendly message shall be displayed. • If a user attempts to access restricted features (e.g., chat) without login, the system shall redirect to the login page. • If internet connectivity is lost, an appropriate error message shall be shown.
Post Conditions	The homepage is successfully displayed to the user. • The user can navigate to product listing pages. • The user can initiate login or registration. • User session activity is logged securely. • No unauthorized access is granted

3.1.2 Buyer Registration

The Buyer Registration module allows new users to create an account on the Kerala Mart website. Registration is required for buyers to initiate chat with sellers, manage their profile, and view chat history. The system ensures secure authentication through validation and OTP verification.

Story ID	KM002
User Story Name	Buyer Registration
As a	New User / Buyer
I want to	Register an account
So that	I can contact sellers and manage my interactions
Pre-Conditions	• The user is on the Homepage. • The user clicks on “Register” under Buyer Login.
Acceptance Criteria	• The system shall display a Buyer Registration form. • The form shall include the following mandatory fields: • Full Name • Email Address • Mobile Number • Password • Confirm Password • The system shall validate: • Email format • Mobile number format • Password strength • Matching passwords • The system shall prevent duplicate registrations using the same email or mobile number. • The system shall send OTP to the registered mobile number or email. • The system shall activate the account only after successful OTP verification. • The system shall securely store user credentials using encryption.
Alternate Flow	• If email already exists, system shall display an appropriate error message. • If OTP is incorrect, system shall display validation error.

- If OTP expires, user shall be able to request resend OTP.
- If required fields are left empty, system shall prompt user to complete the form.

Post Conditions

- Buyer account is successfully created.
- Buyer is automatically logged in (optional) or redirected to login page.
- Registration activity is logged in the system for monitoring and analytics.

3.1.3 Buyer Login

The Buyer Login module allows registered buyers to securely access their Kerala Mart account. Login is required for initiating chat with sellers, viewing chat history, and managing profile settings.

Story ID	KM003
User Story Name	Buyer Login
As a	Registered Buyer
I want to	Log into my account
So that	I can contact sellers and manage my profile
Pre-Conditions	• Buyer account is already registered and verified. • The user is on the Login page.
Acceptance Criteria	• The system shall display a login form. • The form shall require Email or Mobile Number and Password. • The system shall authenticate entered credentials. • The system shall allow a “Forgot Password” option. • The system shall create a secure session upon successful login. • The system shall redirect the buyer to the Homepage after login. • The system shall restrict access if the account is blocked or suspended.

	• The system shall implement session timeout after defined inactivity period.
Alternate Flow	• If incorrect credentials are entered, the system shall display an error message. • If the account is not verified, login shall be restricted. • If password is forgotten, the user shall be redirected to password recovery process.
Post Conditions	• Buyer is successfully logged into the system. • Buyer session is securely established. • Buyer can access chat and profile features.

3.1.4 Forgot Password

The Forgot Password module allows registered buyers to reset their password securely in case they forget their login credentials.

Story ID	KM004
User Story Name	Forgot Password
As a	Registered Buyer
I want to	Reset my password
So that	I can regain access to my account
Pre-Conditions	• Buyer clicks on “Forgot Password” link on Login page.
Acceptance Criteria	• The system shall request registered Email or Mobile Number. • The system shall send OTP to the registered contact method. • The system shall allow user to enter new password after OTP verification.

	• The system shall enforce password strength validation. • The system shall update password securely in database.
Alternate Flow	• If OTP is incorrect, system shall display error message. • If email/mobile not registered, system shall notify user. • If OTP expires, user shall be allowed to resend OTP.
Post Conditions	• Buyer password is successfully reset. • Buyer can log in using new password.

3.1.5 Search & Filter Products

The Search and Filter module allows buyers to easily find vegetables, fruits, and spices by entering keywords and applying filtering criteria. This feature enhances user experience by enabling quick and refined product discovery on the Kerala Mart website.

Story ID	KM006
User Story Name	Search & Filter Product
As a	Buyer
I want to	Search and apply filters to product listings
So that	I can quickly find products that match my needs
Pre-Conditions	• Buyer is on the Homepage or Product Listing Page.
Acceptance Criteria	• The system shall provide a visible search bar in the header. • Buyer shall be able to enter product name (e.g., Tomato, Mango, Pepper). • The system shall support partial keyword matching. • The system shall display matching results based on keyword search. • The system shall provide filter options including: • Category (Vegetables / Fruits / Spices) • Price Range

	• Seller Location • Availability Status • Buyer shall be able to apply one or multiple filters simultaneously. • The system shall dynamically update product results based on selected filters. • The system shall provide an option to clear or reset filters. • If no results match search or filter criteria, system shall display “No Products Found.”
Alternate Flow	• If search field is empty, system shall prompt user to enter keyword. • If applied filters return no results, system shall display appropriate message. • If network issue occurs, system shall show error notification.
Post Conditions	• Relevant filtered product listings are displayed. • Buyer can proceed to view product details. • Buyer can initiate chat with seller.

3.1.6 Product Categories

The Product Categories module allows buyers to browse products based on structured classifications such as Vegetables, Fruits, and Spices. This feature helps users quickly navigate and discover products listed by sellers on the Kerala Mart website.

Story ID	KM006
User Story Name	View Product Categories
As a	Buyer

I want to	Browse products by category
So that	I can easily find vegetables, fruits, or spices
Pre-Conditions	• The user is on the Homepage. • Categories are configured in the Admin panel.
Acceptance Criteria	• The system shall display product categories on the Homepage. • Categories shall include (configurable by Admin): • Fresh Vegetables • Fresh Fruits • Spices • Organic Produce • Each category shall be clickable. • Clicking a category shall redirect to the Product Listing Page. • The system shall display only active categories. • The system shall allow Admin to add, edit, or disable categories.
Alternate Flow	• If no products exist under a category, the system shall display “No Products Available.” • If category is disabled by Admin, it shall not appear on Homepage.
Post Conditions	• Buyer is redirected to the selected category listing page. • Relevant products under that category are displayed.

3.1.7 Product Listing Page

The Product Listing Page displays all products under a selected category or search result. It allows buyers to compare available vegetables, fruits, and spices listed by sellers.

Story ID	KM007
User Story Name	View Product Listings
As a	Buyer
I want to	View a list of available products
So that	I can compare and select products of interest
Pre-Conditions	• Buyer selects a category or performs a search.
Acceptance Criteria	• The system shall display product listings in grid or list format. • Each product card shall include: • Product Image • Product Name • Price (Display Only, e.g., per kg) • Seller Name • Seller Location • “Chat with Seller” button • The system shall support pagination or infinite scroll. • The system shall allow sorting by: • Price (Low to High) • Price (High to Low) • Latest Listings • The system shall not display Cart or Checkout options.
Alternate Flow	• If no products match the criteria, the system shall display “No Results Found.”
Post Conditions	• Buyer can select a product to view detailed information.

3.1.8 Product Detail Page

The Product Detail Page (PDP) provides comprehensive information about a selected vegetable, fruit, or spice listed by a seller. This page enables buyers to review product specifications and directly initiate communication with the seller.

Story ID	KM008
User Story Name	View Product Details
As a	Buyer
I want to	View detailed information about a product
So that	I can make an informed decision before contacting the seller
Pre-Conditions	• Buyer selects a product from the Product Listing Page.
Acceptance Criteria	• The system shall display multiple product images (if uploaded). • The system shall display product name clearly. • The system shall display product price (Display Only, e.g., per kg / per unit). • The system shall display product description. • The system shall display seller name and location. • The system shall display GST Verified badge (if applicable). • The system shall display availability status (Available / Out of Stock). • The system shall provide a visible “Chat with Seller” button. • The system shall not display Add to Cart or Checkout options.
Alternate Flow	• If the product becomes unavailable, the system shall display “Out of Stock.” • If seller account is suspended, the product shall not be accessible. • If product is removed by Admin, the page shall show “Product Not Available.”
Post Conditions	• Buyer can initiate chat with the seller. • Buyer can navigate back to listing page.

3.1.9 Chat System

The Chat module enables real-time communication between buyers and sellers. This feature replaces traditional checkout functionality and allows buyers to negotiate, confirm availability, and finalize transactions offline.

Story ID	KM009
User Story Name	Initiate Chat system
As a	Buyer
I want to	Contact the seller directly
So that	I can discuss price, quantity, and availability
Pre-Conditions	• Buyer must be logged in. • Product detail page is open.
Acceptance Criteria	• The system shall open a chat window when “Chat with Seller” is clicked. • The system shall display seller name in chat header. • The system shall allow text message exchange. • The system shall display timestamp for each message. • The system shall store chat history. • The system shall notify seller of new message. • The system shall allow seller to respond from Seller Dashboard. • The system shall encrypt chat communication.
Alternate Flow	• If buyer is not logged in, system shall redirect to Login page. • If seller is offline, system shall allow message to be sent and delivered later. • If chat service fails, system shall display error message.
Post Conditions	• Chat session is successfully created. • Conversation history is stored in system database.

3.1.10 Buyer Profile Management

The Buyer Profile Management module allows registered buyers to manage their personal information, view chat history, update credentials, and control their account settings within the Kerala Mart website.

Story ID	KM010
User Story Name	Manage Buyer Profile
As a	Logged-in Buyer
I want to	View and update my profile details
So that	I can maintain accurate account information
Pre-Conditions	• Buyer must be logged into the system.
Acceptance Criteria	• The system shall display buyer profile information including: • Full Name • Email Address • Mobile Number • The buyer shall be able to edit profile details. • The system shall validate updated information before saving. • The buyer shall be able to change password. • The buyer shall be able to view chat history. • The buyer shall be able to view account activity (optional).

	• The system shall securely update profile data in database.
Alternate Flow	• If invalid data is entered, the system shall display validation errors. • If password change fails due to incorrect current password, system shall notify the user. • If session expires during update, user shall be redirected to login page.
Post Conditions	• Profile information is successfully updated. • Changes are saved securely in system. • User session remains active (unless password reset requires re-login).

3.1.11 Notifications

The Notifications module allows buyers to receive important updates related to their account activity, including chat messages, product availability updates, and system announcements. This feature ensures buyers remain informed about interactions and platform updates.

Story ID	KM011
User Story Name	Notifications
As a	Buyer
I want to	Receive notifications
So that	I can stay updated on important activities
Pre-Conditions	• User is registered and logged into the system.
Acceptance Criteria	• The system shall provide a notification icon in the header. • The system shall display unread notification count. • The system shall allow users to click and view notification details. • Notifications shall include timestamp information. • The system shall store notification history.

Buyer Notification Types

- New message received from seller.
- Seller response to inquiry.
- Account updates (profile changes, password reset).

Alternate Flow	• If notification service temporarily fails → system shall still store events and display them once service resumes. • If user clears notifications → system shall mark them as read.
Post Conditions	• User receives timely alerts. • Notification history is stored securely.

3.1.12 Buyer Account Settings

The Buyer Account Settings module allows buyers to manage personal account information, security preferences, and account controls.

Story ID	KM013
User Story Name	User Story Name
As a	Buyer
I want to	Manage my account settings
So that	I can maintain my personal information and security
Pre-Conditions	• Buyer is logged into the system.
Acceptance Criteria	• Buyer shall be able to view and update:

	• Full Name • Email Address • Mobile Number
	• Buyer shall be able to change password. • Buyer shall be able to enable/disable notification preferences (if configurable). • Buyer shall be able to delete or deactivate account. • The system shall validate all updates before saving. • The system shall securely store updated information.
Alternate Flow	• If invalid data is entered → system shall display validation error. • If password confirmation does not match → system shall display error. • If session expires during update → user shall be redirected to login page.
Post Conditions	• Buyer account settings are successfully updated. • Updated information is reflected in system database.

3.1.13 Delete Buyer Account

The Delete Account module allows buyers to permanently deactivate or delete their Kerala Mart account if they no longer wish to use the platform.

Story ID	KM011
User Story Name	Delete Buyer Account
As a	Registered Buyer
I want to	Delete my account
So that	I can permanently remove my data from the platform

Pre-Conditions	• Buyer must be logged in. • Buyer accesses account settings page.
Acceptance Criteria	• The system shall provide an option to delete account. • The system shall request confirmation before deletion. • The system shall require password verification before deletion. • The system shall deactivate buyer account upon confirmation. • The system shall archive chat history for compliance purposes (if required). • The system shall prevent login after deletion.
Alternate Flow	• If password verification fails, system shall deny deletion request. • If buyer cancels deletion request, account remains active.
Post Conditions	• Buyer account is permanently deactivated. • Buyer data is handled according to privacy policy. • Buyer can no longer access the platform.

3.2 Seller Functionalities

3.2.1 Seller Registration

The Seller Registration module allows farmers, vegetable vendors, fruit suppliers, and spice traders to create an account on the Kerala Mart website. Seller registration requires mandatory GST verification and Admin approval before activation.

Story ID	KM101
User Story Name	Seller Registration
As a	Business Seller
I want to	Register my business on Kerala Mart
So that	I can list my products
Pre-Conditions	• Seller clicks on “Sell on Kerala Mart” CTA button. • Seller is redirected to Seller Registration page.
Acceptance Criteria	• The system shall display Seller Registration form. • The form shall include mandatory fields: • Business Name • Owner Name • Email Address • Mobile Number • Business Address • GST Number (Mandatory) • GST Certificate Upload • Password • The system shall validate GST number format. • The system shall allow upload of GST certificate (PDF/JPEG). • The system shall validate email and mobile number format. • The system shall prevent duplicate registration.

	• The system shall mark seller status as “Pending Approval.” • Seller account shall remain inactive until Admin approval.
Alternate Flow	• If GST number format is invalid, system shall show validation error. • If GST document is not uploaded, registration shall not proceed. • If email already exists, system shall display error.
Post Conditions	• Seller account is successfully created with “Pending” status. • Admin is notified for verification.

3.2.2 Seller Login

The Seller Login module allows approved sellers to securely access their Seller Dashboard.

Story ID	KM102
User Story Name	Seller Login
As a	Approved Seller
I want to	Log into my account
So that	I can manage my listings
Pre-Conditions	• Seller account is registered. • Seller account is approved by Admin.

Acceptance Criteria	• The system shall display a Seller Login page. • Seller shall enter registered Email or Mobile Number and Password. • The system shall authenticate the entered credentials. • The system shall redirect the seller to the Seller Dashboard upon successful login. • The system shall restrict login if the seller account is inactive or suspended. • The system shall provide a “Forgot Password” option. • The system shall implement session timeout after inactivity.
Alternate Flow	• If incorrect credentials are entered → system shall display “Invalid Credentials.” • If seller account is not approved → system shall display “Account Pending Approval.” • If account is suspended → system shall display appropriate message.
Post Conditions	• Seller session is successfully initiated. • Seller dashboard becomes accessible.

3.2.3 Seller Dashboard

The Seller Dashboard is the central control panel where sellers manage their products, view subscription details, and respond to buyer inquiries.

Story ID	KM104
User Story Name	Seller Dashboard
As a	Seller
I want to	Edit or delete my product listings
So that	I can manage my activities efficiently
Pre-Conditions	• Seller is logged in.
Acceptance Criteria	• Dashboard shall display:

	• Total Active Listings • Total Inquiries • Subscription Status • Expiry Date
	• Seller shall have options to: • Add Product • Manage Products • View Chats • Edit Profile • Dashboard shall update data in real-time or near real-time.
Alternate Flow	• If subscription expired → system shall display renewal alert.
Post Conditions	• Seller can navigate to different management modules.

3.2.4 Purchase Subscription Plan

This module allows sellers to purchase a subscription plan to activate product listing functionality on the website.

Story ID	KM103
User Story Name	Purchase Subscription
As a	Seller
I want to	Subscribe to a listing plan
So that	I can publish my product listings
Pre-Conditions	• Seller is logged in. • Seller account is approved.

Acceptance Criteria	• The system shall display available subscription plans (Monthly / Quarterly / Annual). • Each plan shall display price and duration clearly. • Seller shall select a subscription plan. • Seller shall complete payment via payment gateway. • Upon successful payment: • Subscription status shall change to “Active.” • Subscription expiry date shall be displayed. • The system shall send confirmation notification after payment. • The system shall automatically deactivate listings if subscription expires.
Alternate Flow	• If payment fails → subscription shall not activate. • If seller cancels payment → system returns to plan selection page.
Post Conditions	• Seller subscription is active. • Seller can now add product listings

3.2.5 Add Product

Allows sellers to list vegetables, fruits, and spices on the Kerala Mart website.

Story ID	KM105
User Story Name	Add Product
As a	Seller
I want to	Add product details
So that	Buyers can view and contact me
Pre-Conditions	• Seller is logged in. • Subscription is active.

Acceptance Criteria	• Seller shall enter: • Product Name • Category • Price (per kg / per unit) • Description • Availability Status • Upload product images • System shall validate required fields. • Product shall be visible after submission.
Alternate Flow	• If subscription expired → listing not allowed. • Missing fields → validation error.
Post Conditions	• Product is successfully listed.

3.2.6 Manage Products

The Manage Products module allows sellers to view, update, and remove their listed vegetables, fruits, and spices from the Kerala Mart website. This functionality ensures that product information remains accurate, up-to-date, and relevant for buyers.

Story ID	KM106
User Story Name	Manage Product Listings
As a	Seller
I want to	Edit, update, or delete my listed products
So that	My product information remains accurate and current
Pre-Conditions	• Seller is logged into the system. • Seller has an active subscription. • Seller has at least one product listed.

Acceptance Criteria

- The system shall display a list of all products added by the seller in the Seller Dashboard.
- Each product entry shall display:
 - Product Name
 - Category
 - Price (per kg / per unit)
 - Availability Status
 - Date Added
 - Edit option
 - Delete option
- Seller shall be able to click “Edit” to modify:
 - Product Name
 - Price
 - Description
 - Availability Status
 - Product Images
- The system shall validate updated information before saving changes.
- Updated product details shall reflect immediately on the website.
- Seller shall be able to change product availability status to:
 - Available
 - Out of Stock
- Seller shall be able to delete a product listing.
- Before deletion, the system shall request confirmation from the seller.
- Upon deletion, the product shall be removed from public view.
- The system shall log product modification activities for audit purposes.

Alternate Flow	• If subscription has expired → system shall allow editing existing products but restrict adding new products (based on business rule). • If required fields are left empty during editing → system shall display validation error. • If seller cancels deletion confirmation → product shall remain active. • If network failure occurs during update → system shall display error message and prevent data loss.
Post Conditions	• Product details are successfully updated in the system database. • Deleted products are removed from buyer view. • Dashboard reflects updated product count and status. • Seller product records remain consistent and synchronized with system data.

3.2.7 Chat

The Respond to Buyer Chat module enables sellers to communicate directly with buyers who have shown interest in their listed vegetables, fruits, or spices. This feature supports negotiation, quantity confirmation, price clarification, and finalization of transactions outside the platform.

Story ID	KM107
User Story Name	Respond to Buyer Inquiry
As a	Seller
I want to	View and respond to buyer messages

So that	I can provide product details and finalize sales
Pre-Conditions	• Seller is logged into the system. • Seller has at least one active product listing. • Buyer has initiated chat from Product Detail Page.
Acceptance Criteria	• The system shall notify the seller when a new message is received. • The system shall display unread message count on the dashboard. • The seller shall be able to open chat from: • Seller Dashboard • Notification panel • The chat interface shall display: • Buyer name • Product name (related to inquiry) • Message history • Timestamp for each message • Seller shall be able to: • Send text messages • View previous messages • Respond in real-time • The system shall store complete chat history securely. • The system shall ensure chat messages are encrypted during transmission. • Seller shall receive notification for each new incoming message. • The chat window shall auto-refresh or update in near real-time.
Alternate Flow	• If buyer is offline → seller can still send message, and buyer shall receive it upon next login.

- If network connectivity is interrupted → system shall display connection error and retry option.
- If seller account is suspended → chat functionality shall be restricted.
- If buyer account is deleted → chat history shall remain archived for record purposes.

Post Conditions

- Seller and buyer communication is successfully recorded.
- Chat history is stored in system database.
- Seller can continue conversation anytime.
- System logs chat activity for monitoring and compliance purposes.

3.2.8 Seller Profile Management

The Seller Profile Management module allows sellers to view and update their business information, contact details, and account settings on the Kerala Mart website. This ensures that seller information remains accurate and trustworthy for buyers.

Story ID	KM108
User Story Name	Manage Seller Profile
As a	Seller
I want to	View and update my business profile
So that	Buyers can see accurate business information
Pre-Conditions	• Seller is logged into the system.
Acceptance Criteria	• The system shall display seller profile details including: • Business Name

- Owner Name
- Email Address
- Mobile Number
- Business Address
- GST Number (view-only)
- Seller shall be able to edit:
 - Business Name
 - Contact Number
 - Business Address
 - Profile Image (optional)
- Seller shall be able to change password.
- The system shall validate updated information before saving.
- Updated profile information shall reflect immediately in:
 - Seller Dashboard
 - Product Detail Page (Seller Info Section)
- The system shall securely store updated data.

Alternate Flow

- If invalid data is entered → system shall display validation error.
- If password change fails due to incorrect current password → system shall display error message.
- If session expires during update → seller shall be redirected to login page.

Post Conditions

- Seller profile information is successfully updated.
- Updated information is reflected across the website.
- System logs profile update activity.

3.2.9 Seller Notifications

The Seller Notification module ensures that sellers receive timely alerts related to buyer inquiries, subscription status, and account updates.

Story ID	KM109
User Story Name	Seller Notification
As a	Seller
I want to	Receive notifications
So that	I can respond promptly to buyer inquiries and manage my subscription
Pre-Conditions	• Seller is logged into the system.
Acceptance Criteria	• The system shall display a notification icon in the header. • The system shall show unread notification count. • Seller shall receive notifications for: • New buyer message • New inquiry • Subscription expiry reminder • Subscription activation confirmation • Account status updates • Seller shall be able to click and view notification details. • Notifications shall display timestamp. • Notification history shall be stored securely.
Alternate Flow	• If notification service fails temporarily → system shall store events and display them once restored. • If seller clears notification → system shall mark it as read.
Post Conditions	• Seller remains informed about important system activities.

- Notification history remains accessible.

3.2.10 Subscription Expiry & Renewal

The Subscription Expiry & Renewal module allows sellers to monitor subscription status, receive expiry alerts, and renew their subscription to continue listing products on the platform..

Story ID	KM110
User Story Name	Subscription Expiry & Renewal
As a	Seller
I want to	Renew my subscription before expiry
So that	My product listings remain active
Pre-Conditions	• Seller is logged in. • Seller has an active or expiring subscription.
Acceptance Criteria	• System shall display subscription status in Seller Dashboard. • System shall display: • Subscription Plan Name • Expiry Date • Days Remaining • System shall send notification reminders: • Before 7 days of expiry • On expiry date • Seller shall be able to click “Renew Subscription” button. • System shall redirect seller to subscription plan selection page.

	• Upon successful payment: • Subscription status shall update to Active • New expiry date shall be calculated • If subscription expires: • System shall restrict new product additions • Existing listings may be hidden (based on business rule)
Alternate Flow	• If payment fails → subscription shall remain inactive. • If seller ignores renewal → system shall automatically deactivate listing capability. • If seller renews before expiry → continuity shall be maintained without service interruption.
Post Conditions	• Seller subscription is renewed successfully. • Product listing capability remains active. • Seller dashboard reflects updated subscription details.

4. Admin Panel Features

The Admin Panel is a secure backend management interface designed exclusively for authorized administrators of the Kerala Mart website. It provides centralized control over user management, seller verification, subscription configuration, product moderation, and operational monitoring.

4.1 Admin Login

The Admin Login module allows authorized administrators to securely access the Kerala Mart Admin Panel. This module ensures that only authenticated and authorized personnel can manage platform operations such as seller approval, subscription management, product moderation, and system monitoring.

The login process shall implement strong security mechanisms to protect sensitive platform data and prevent unauthorized access.

Sections Included

- **Secure Login Form:**

Provides a protected interface where Admin enters credentials to access the system.

- **Username / Email Input:**

Allows Admin to enter registered username or email for identification.

- **Password Input:**

Allows Admin to enter secure password, which is encrypted and validated by the system.

- **Forgot Password Option:**

Enables Admin to reset password securely if forgotten.

- **Multi-level Authentication (if implemented):**

Adds an additional security layer such as OTP verification.

- **Session Timeout Control:**

Automatically logs out Admin after a period of inactivity for security.

- **Login Attempt Monitoring:**

Tracks login attempts to detect suspicious activities.

- **Account Lock Mechanism:**

Temporarily locks account after multiple failed login attempts to prevent unauthorized access.

4.2 Seller Management

The Seller Management module allows the Admin to review, approve, monitor, and control seller accounts on the Kerala Mart platform.

Sections Included

- **View Seller List:**

Displays all registered sellers in the system.

- **Search & Filter Sellers:**

Allows Admin to search sellers by Business Name, Email, or GST Number and filter by account status.

- **View Seller Details:**

Shows complete seller profile including business information and GST details.

- **Approve Seller Registration:**

Allows Admin to activate seller accounts after verification.

- **Reject Seller Registration:**

Allows Admin to deny registration if details are invalid.

- **Suspend Seller Account:**

Temporarily restricts seller access for policy violations.

- **Reactivate Seller Account:**

Restores seller access after suspension.

- **View Seller Subscription Status:**

Displays active plan, expiry date, and payment details.

- **Seller Activity Monitoring:**

Allows Admin to view seller's listed products and activity logs.

4.3 Subscription Plan Management

The Subscription Plan Management module allows the Admin to create, update, and manage subscription plans that sellers must purchase to list products on the platform.

Sections Included

- **Create Subscription Plan:**

Allows Admin to define new plans with name, duration, and price.

- **Edit Subscription Plan:**

Enables modification of plan pricing, duration, or description.

- **Activate / Deactivate Plan:**

Allows Admin to make plans available or unavailable for sellers.

- **Set Plan Duration & Price:**

Defines how long the subscription is valid and how much it costs.

- **Monitor Active Subscriptions:**

Displays sellers currently subscribed to each plan.

- **View Revenue Summary:**

Shows total revenue generated from subscriptions.

- **Expiry Monitoring:**

Allows Admin to track expiring subscriptions.

4.4 Product Moderation

The Product Moderation module allows the Admin to monitor and manage product listings to ensure quality, accuracy, and compliance with platform policies.

Sections Included

- **View All Product Listings:**

Displays all vegetables, fruits, and spices listed by sellers.

- **Search & Filter Products:**

Allows Admin to filter products by category, seller, or status.

- **View Product Details:**

Shows product information including images, price, description, and seller details.

- **Approve Product (if moderation enabled):**

Allows Admin to approve product before it becomes visible to buyers.

- **Remove Product Listing:**

Allows Admin to delete inappropriate or policy-violating products.

- **Disable Product Temporarily:**

Temporarily hides product without permanent deletion.

- **Moderation History Log:**

Maintains record of all product-related admin actions.

5. Non Functional Requirements

Non-functional requirements for the Multi-vendor marketplace platform PWA and Mobile app can include the following:

Feature	Description
Security	The website should be safe, with safeguards in place to fend off viruses, hacking, and other online dangers. Firewalls, SSL encryption, frequent updates, and user authentication are a few examples of such features.
Performance	The website must load swiftly and react to user queries promptly. Image optimization, a reduction in HTTP requests, and the use of caching plugins can all help to speed up page load times.
Scalability	The website should be able to handle increasing traffic and user numbers without experiencing downtime or performance problems. This may entail using a cloudbased hosting provider, load balancing, and resource scaling as necessary.
Usability	The website should have simple navigation, unambiguous labeling, and a unified look and feel. This may entail carrying out user research, putting responsive design into practice, and offering accessibility features to users with disabilities.

SEO	With elements like meta tags, keyword optimization, and high-quality content that is updated frequently, the website should be search engine optimized. This could increase the website's visibility and draw in more visitors.
Maintenance & Support	The website should be simple to update and maintain, with a support staff on hand to address any potential technical problems. Version control, regular backups, and training for website Admin can all be part of this.
Compatibility	A wide range of browsers and devices, including desktop , should be able to access the website. This may entail testing the website across various platforms and implementing responsive design to guarantee that it functions properly across all gadgets.
Availability	There shouldn't be much downtime or maintenance scheduled for the website. This may entail putting in place redundancy and failover mechanisms as well as monitoring tools to make sure the website is constantly operational.

6. System Integrations

The following integrations are considered for the proposed system.

Integration	Specification
Product & Inventory Update	To be Confirmed
Advanced Search	To be Confirmed
SMS Gateway	To be Confirmed
Email Gateway	To be Confirmed
Payment Gateway	To be Confirmed
Shipping Gateway	To be Confirmed
Buy Now Pay Later	To be Confirmed
Chatbot	To be Confirmed
Social Logins	To be Confirmed
Omni Channel Integration	To be Confirmed
Web Analytics	To be Confirmed
Whatsapp Notification	To be Confirmed

7. Assumptions & Dependencies

A website's assumptions and dependencies refer to the underlying conditions and factors required for the successful development and operation of the website.

- **Assumptions**

- To access the Multi-vendor marketplace platform, customers must have a stable internet connection and compatible devices (such as smartphones, tablets, or computers).
- The system keeps the product catalog and inventory up to date and accurate.
- The system can handle a reasonable number of concurrent users and transactions without experiencing performance problems.
- The payment gateway integration will be successfully installed and configured with the appropriate security measures.
- Customers will understand the fundamentals of navigating and using an e-commerce website.
- The system will adhere to all applicable legal and regulatory requirements, such as data protection and consumer rights.

- **Dependencies**

- **Technology Stack:** The availability and proper configuration of the chosen technology stack, such as Flutter and MERN, is critical to the system's successful implementation. It is critical to ensure the compatibility, version compatibility, and proper installation of these technologies.
- **Third-Party APIs and Integrations:** If the system needs to integrate with third-party services such as payment gateways, shipping providers, or analytics tools, the availability and proper operation of these APIs is critical. Access to and integration with these APIs should be accompanied by documentation and credentials from the client side.
- **Database Management:** The system stores and retrieves data using a database management system . Proper installation, configuration, and database system access are required dependencies.
- **Design Assets:** The development team relies on the design team or stakeholders to provide design assets such as wireframes, UI/UX designs, and branding elements. These assets will guide the user interface's development and ensure a consistent visual identity.
- **Product Data and Content:** Implementing the product listing and detail pages requires the availability of accurate and up-to-date product data, such as descriptions, images, prices, and specifications. The client team should provide this information in a format suitable for system integration.

- Testing Environment: It is critical for testing and quality assurance to have a separate testing environment from the production environment. Access to and provisioning of the testing environment should be ensured in order to conduct comprehensive testing of the system's functionality and performance.

8. Conclusion

This project involved the preparation of a comprehensive **Software Requirements Specification (SRS)** for the *Kerala Mart B2B Marketplace Platform* as part of my internship at **Zoople Technologies**. The objective was to apply Business Analysis practices in a real-world scenario by transforming business needs into structured, professional documentation aligned with industry standards.

During the course of this project, the complete system was analyzed from multiple perspectives including business goals, user roles, functional requirements, non-functional requirements, and system integrations. A detailed set of user stories, workflows, and platform modules were defined to clearly describe how buyers, suppliers, affiliates, and administrators interact with the platform. This documentation ensures that all stakeholders share a common understanding of the system's expected behavior before development begins.

The project provided valuable hands-on experience in:

- Requirements gathering and analysis
- Writing professional SRS documentation
- Creating structured user stories and acceptance criteria
- Understanding B2B marketplace workflows
- Applying software engineering and Agile documentation practices

By completing this project, I gained practical exposure to the role of a **Business Analyst** and learned how structured documentation plays a critical role in reducing ambiguity, improving collaboration, and guiding successful product development.

Overall, this internship project successfully demonstrates the ability to translate business ideas into clear, scalable, and implementation-ready requirements, making it a strong foundation for future professional work in software development and business analysis.

9. Disclaimer

This Software Requirements Specification (SRS) document has been prepared as part of an academic internship project at **Zoople Technologies** and is intended solely for educational and documentation purposes. Any reproduction, distribution, or use of this document without prior permission from the author is not permitted.

The features, workflows, and technical details described in this document represent a proposed system based on research and analysis and may vary during actual development depending on stakeholder requirements, technical constraints, and business decisions.

All information presented in this document should be considered a conceptual and academic representation of the proposed platform. The reader is advised to independently validate the information before using it for commercial or development purposes.